

Independent Study Report

Automated Species-Level Classification of Caribbean Coral Reefs

A SAM2 Segmentation and EfficientNet-B3 Pipeline

Mitanshu Mothghare

May 2026

Abstract

This report documents an end-to-end deep learning pipeline for identifying Caribbean coral species from underwater photoquadrat imagery. The system pairs the second-generation Segment Anything Model (SAM2) for instance-level coral colony segmentation with a fine-tuned EfficientNet-B3 convolutional network for species classification across 17 Caribbean scleractinian and hydrozoan species. Training data was assembled from eight CoralNet sources spanning reef sites in Curaçao, Bonaire, Belize, Little Cayman, and St. John, producing 66,219 species-level annotations across 19,152 underwater images.

A central methodological decision in this work was to generate training patches using SAM2 *point prompts* rather than automatic mask generation. Each CoralNet annotation point became a foreground prompt to SAM2, yielding a colony mask with a guaranteed species label and avoiding the false positives endemic to automatic segmentation. The final model achieves a test macro-F1 of 0.624 and overall accuracy of 72.6%, surpassing a comparably trained MobileSAM-based baseline (macro-F1 = 0.606). A negative result is also reported: substituting EfficientNet-B3 with the self-supervised Vision Transformer DINOv2-S degraded macro-F1 to 0.376, almost certainly because the zeroed background of the masked crops violates the input distribution under which DINOv2 was pretrained. Persistent failure modes include severe class imbalance (the rarest class, *Meandrina meandrites*, has only 378 training patches against 7,226 for *Madracis mirabilis*) and morphology-specific under-segmentation for thin branching and small foliate colonies.

The pipeline is fully implemented and publicly released; the report documents the experimental choices, compares them against published baselines, and identifies concrete next steps for continued development. All code is available at <https://github.com/ARLab2018/Caribbean-Coral-Classification>, with trained model weights, source images, and patches hosted on Google Drive (links in the repository).

Contents

1	Introduction	4
1.1	Scope and contributions	4
2	Background and Related Work	5
2.1	The Segment Anything family	5
2.2	Coral-specific segmentation models	7
2.3	Vision Transformers and classification backbones	8
2.4	Class imbalance and long-tailed learning	9
2.5	Pre-foundation-model coral classification	10
3	Dataset and Annotation	10
3.1	Species mapping	11
3.2	Train, validation, and test splits	12
4	Methodology	12
4.1	Pipeline overview	12
4.2	Segmentation: SAM2	13
4.3	Patch generation with SAM2 point prompts	13
4.4	Classification: EfficientNet-B3	14
5	Experiments and Results	15
5.1	Segmentation model: MobileSAM versus SAM2	15
5.2	Classification backbone: EfficientNet-B3 versus DINOv2-S	16
5.3	Final model: per-species results	16
5.4	Patch-count comparison	17
6	Discussion	18
6.1	Why SAM2 helps	18
6.2	Why DINOv2 fails (and what to do about it)	19
6.3	Persistent limitations	20
6.4	Comparison to published baselines	20
7	Conclusions and Future Work	21
7.1	Recommendations for follow-on work	22
7.2	Closing remarks	22

1 Introduction

Coral reefs support a disproportionately large fraction of marine biodiversity relative to the area of seafloor they occupy, and the goods and services they provide — coastal protection, fisheries productivity, tourism revenue — are economically significant well beyond the reef itself (Eddy et al., 2021). The problem, of course, is that these ecosystems are in serious trouble. A recent global synthesis concluded that living coral cover has fallen by roughly 50% since the 1950s, with catch-per-unit-effort for reef-associated fishes down by about 60% over the same period (Eddy et al., 2021). The Caribbean has fared particularly badly: cover here declined by more than half between the 1970s and 2012, with the loss of grazing parrotfish populations identified as a major proximate driver (Jackson et al., 2014).

Long-term monitoring is the foundation on which any management response must rest, but the dominant methods — diver-led line-intercept transects and the photoquadrat-with-point-annotations workflow popularised by CoralNet (Beijbom et al., 2015) — are slow, labour-intensive, and notoriously inconsistent across observers. Beijbom et al. reported that even trained experts agree on coral identity at the genus level but diverge sharply when asked to discriminate between functional algae groups such as turf and crustose coralline algae. Automated classification, if it can be made reliable, addresses both the labour problem and the consistency problem at once.

The last two years have seen the field shift dramatically. The release of the Segment Anything Model (Kirillov et al., 2023) demonstrated that a single segmentation network, trained at scale on diverse imagery, could be prompted to produce competent masks for almost any object class without further task-specific training. Its successor, SAM2 (Ravi et al., 2024), extended this capability to video and improved efficiency by an order of magnitude. Specialised derivatives have followed quickly: CoralSCOP (Zheng et al., 2024a) fine-tuned SAM on more than 41,000 coral images and 330,000 masks, providing the first coral-aware foundation model for dense reef segmentation. The opportunity that motivates this study is straightforward: combine the strong instance segmentation of a foundation model with a focused species classifier and see what level of species-level accuracy is achievable across the species commonly encountered in Caribbean photoquadrats.

Over the course of the project I built and evaluated four end-to-end configurations: MobileSAM (Zhang et al., 2023a) paired with EfficientNet-B3, CoralSCOP (Zheng et al., 2024a) paired with EfficientNet-B3, SAM2 (Ravi et al., 2024) paired with EfficientNet-B3, and SAM2 paired with the self-supervised Vision Transformer DINOv2-S (Oquab et al., 2024). The first three were taken to convergence; the CoralSCOP-based classifier did not train successfully and is included as a documented negative outcome. The third configuration — SAM2 with EfficientNet-B3 — is the final pipeline this report defends.

1.1 Scope and contributions

This report describes the design, training, and evaluation of one such pipeline. The contributions are modest but, I think, useful for guiding follow-on work:

- A complete pipeline from raw underwater images to species-labelled colony maps, built

around SAM2 for segmentation and EfficientNet-B3 (Tan & Le, 2019) for classification.

- A patch-generation method that uses SAM2 with point prompts rather than automatic mask generation, exploiting the fact that CoralNet annotations are already given as point coordinates with verified species labels. This yields cleaner training data than the automatic-mask approach previously used with MobileSAM (Zhang et al., 2023a), and the resulting classifier improves macro-F1 by 0.018 despite training on roughly 6,000 fewer patches.
- A negative finding on Vision Transformers in this setting: a DINOv2-S backbone (Oquab et al., 2024) fails badly on the same masked crops that EfficientNet-B3 handles competently, dropping macro-F1 by more than 0.2. The most plausible explanation is a distribution mismatch between the zeroed background of masked crops and the natural-image patches DINOv2 was pretrained on, and this in turn suggests that either bounding-box crops with natural background or explicit attention masking would be necessary to make ViTs work here.
- An empirical characterisation of where the pipeline falls short — in particular, the species that share both low annotation counts and visual similarity with congeners (*Orbicella franksi*, *Porites porites*, *Meandrina meandrites*) — alongside concrete suggestions for closing those gaps.

The remainder of this report is structured conventionally. Section 2 reviews relevant work in foundation segmentation models, Vision Transformers, class imbalance methods, and prior automated coral classification systems. Section 3 describes the dataset and labelling. Section 4 details the segmentation and classification stages. Section 5 reports the main experiments and ablation studies. Section 6 interprets the results in relation to the broader literature, and Section 7 closes with the limitations of the current system and recommendations for next steps.

2 Background and Related Work

This section surveys the four bodies of work most relevant to the pipeline described in the report: foundation models for segmentation, Vision Transformer architectures and their use as classification backbones, methods for handling class imbalance in deep learning, and prior automated systems for coral reef image analysis. The aim is not exhaustive coverage but rather to position the present work clearly within ongoing research and to make explicit the assumptions inherited from each line of work.

2.1 The Segment Anything family

The release of the Segment Anything Model (SAM) by Meta AI in early 2023 was, by any measure, a turning point for segmentation research (Kirillov et al., 2023). The system consists of three loosely coupled components: a ViT-H image encoder that produces a dense embedding of the input image, a lightweight prompt encoder that processes either points, boxes, or text, and a fast mask decoder that fuses the two and produces multiple plausible masks for the prompt. The training set — SA-1B, comprising about 1.1 billion masks across 11 million licensed images — was assembled through a three-stage semi-automated pipeline that progressively replaced human annotation with model-generated masks. The result was a model that could produce competent

segmentations of novel objects, in novel domains, with only a single foreground point as a prompt; this generality is what makes "foundation model" a meaningful description in this context.

SAM is not without weaknesses. The original ViT-H encoder is heavy — approximately 632 million parameters, requiring substantial GPU memory and producing latencies that are impractical for edge deployment or real-time video. Mask boundaries on intricate objects, such as branching corals or thin leaves, often appear noticeably smoother than the underlying object. And the model's behaviour on out-of-distribution imagery — including, notably, underwater scenes — is uneven (Lian & Li, 2024). A substantial follow-up literature has emerged addressing these issues.

Efficiency-oriented variants. The first wave of derivative work focused on shrinking SAM's image encoder while preserving mask quality. MobileSAM (Zhang et al., 2023a) replaces the ViT-H encoder with a Tiny-ViT of about five million parameters using a decoupled knowledge-distillation scheme; the resulting model is roughly 60× smaller overall and runs the encoder in around eight milliseconds. FastSAM (Zhao et al., 2023) takes a different route, recasting the segment-anything problem as YOLOv8-based instance segmentation followed by prompt-guided selection, training on only 2% of SA-1B and running about 50× faster than the original SAM on equivalent hardware. EfficientSAM (Xiong et al., 2024) introduces SAMI, a masked image pretraining objective in which a light ViT reconstructs the features of SAM's heavy ViT-H encoder; this delivers roughly 4 AP improvement on COCO zero-shot instance segmentation over comparably sized models. None of these address segmentation quality directly — they trade some mask precision for substantial gains in inference speed and deployability.

Quality-oriented variants. A parallel line of work has pushed in the opposite direction. HQ-SAM (Ke et al., 2023) introduces a learnable High-Quality Output Token and a fused early/late-feature pathway in the mask decoder, trained on a curated set of about 44,000 high-quality masks. With less than 0.5% additional parameters, HQ-SAM improves average boundary IoU by 10.7 points across four difficult benchmarks while keeping SAM's zero-shot generalisation intact. The lesson from this line of work is that SAM's mask quality is bottlenecked less by encoder capacity than by the granularity and consistency of the masks in its training data.

SAM2 and video. The second major release in this lineage is SAM2 (Ravi et al., 2024), which extends the framework to video by introducing a streaming memory module that propagates prompts and mask predictions across frames. The image encoder is replaced with Hiera (Ryali et al., 2023), a hierarchical ViT pretrained with masked autoencoding that delivers higher accuracy than Swin or MViTv2 (Liu et al., 2021) while being faster at inference. The training data for SAM2 includes the new SA-V dataset, which the authors report as containing roughly 50.9 thousand videos and 35.5 million masks. On still-image segmentation, SAM2 reports a 6× speed-up over SAM and requires about 3× fewer interactive prompts to converge on a target object. For my purposes, the relevant features are simply that mask boundaries are sharper on textured objects (such as coral colonies with visible corallite structure) and that the image encoder, while still substantial, is small enough to run alongside the EfficientNet-B3 classifier on an 8 GB consumer GPU.

SAM and SAM2 underwater. Several recent studies have characterised how the SAM family behaves in marine settings. Lian and Li (Lian & Li, 2024) report that SAM2 produces excellent

segmentations underwater when given a ground-truth bounding-box prompt but degrades sharply in fully automatic point-prompt mode. Lian et al. (Lian et al., 2024) introduced USIS10K, a 10,632-image underwater salient instance segmentation dataset with pixel-level annotations across seven categories, alongside a SAM-based architecture adapted to the optical characteristics of underwater imagery. These results informed two design choices in the present pipeline: I avoid relying on pure automatic mask generation at inference time when an alternative is available, and I use point prompts (where I have prior knowledge of where coral exists) rather than dense grid sampling for training-patch extraction.

2.2 Coral-specific segmentation models

The most relevant prior art is CoralSCOP (Zheng et al., 2024a), presented as a CVPR 2024 Highlight. The authors fine-tuned SAM with a parallel semantic branch on a new dataset — CoralMask, containing 41,297 coral images and 330,144 densely labelled masks — and demonstrate convincingly that vanilla SAM over-segments coral imagery, producing many small masks where a single colony exists, whereas CoralSCOP produces coherent, semantically labelled coral masks. The model has been packaged into an offline labelling tool, CoralSCOP-LAT, that targets the workflow of working marine biologists (Wong et al., 2024). CoralSCOP does not provide species-level labels — it operates at the coral-vs-non-coral level — and so it does not directly solve the problem this report addresses, but it is the natural baseline against which a SAM2 plus species classifier pipeline should be compared and the most obvious candidate for a future drop-in replacement for the SAM2 segmentation stage.

Other recent work pushes in adjacent directions. MarineInst (Zheng et al., 2024c), presented at ECCV 2024 as an oral paper, builds a marine-domain foundation model that mitigates SAM’s tendency toward over- and under-segmentation on marine imagery and adds per-instance natural-language descriptions. Coral-Lab (Wang et al., 2025), recently published in IJAEOG, takes a different design philosophy entirely — modifying DeepLabV3+ with an Attention-ASPP module and four-level feature fusion — and achieves F-score 0.802 and mIoU 0.665 on a curated mix of local and external coral imagery, deployed via a towed camera array (SSSV2) for large-scale reef mapping. The Coral-Lab paper is also notable for an empirical finding I return to repeatedly in this work: data *curation* dominates raw data *quantity* in determining downstream performance, with their D7 dataset (970 carefully selected images) outperforming the D6 superset (1,085 images) by an F-score of 0.802 versus 0.735.

For completeness, two older systems deserve mention as historical baselines. Beijbom et al.’s CoralNet (Beijbom et al., 2015) is both the source of most of the training annotations used in this work and the de facto standard reference for automated benthic image annotation. CoralSeg (Alonso et al., 2019) provides the first practical pipeline for converting CoralNet-style sparse point annotations into dense semantic segmentations through adaptive superpixel propagation; the present work can be read as a foundation-model successor to that approach. Gómez-Ríos et al. (Gómez-Ríos et al., 2019) provides a useful pre-foundation-model benchmark, applying ResNet, Inception, and DenseNet variants with aggressive data augmentation to the EILAT and RSMAS coral texture datasets and showing that augmentation can close most — though not all — of the imbalance gap.

2.3 Vision Transformers and classification backbones

For the classification stage of the pipeline, the choice of backbone interacts in non-obvious ways with the structure of the training data, and a brief review of the relevant architectures and their inductive biases is warranted.

From convolutions to attention. Convolutional networks dominated image classification from roughly 2012 to 2020, with ResNet (He et al., 2016) serving as the canonical reference architecture. The introduction of residual connections allowed networks to be scaled to hundreds of layers without optimisation pathologies, and ResNet remains the standard baseline for almost any vision benchmark. EfficientNet (Tan & Le, 2019) subsequently demonstrated that compound scaling of depth, width, and input resolution, found through neural architecture search, produces a family of models that strictly dominate ResNet on the accuracy-versus-FLOPs frontier — EfficientNet-B7 achieved 84.4% top-1 ImageNet accuracy with $8.4\times$ fewer parameters than the best prior ConvNet. EfficientNet-B3, the variant used in this report, is the practical sweet spot for moderate-sized GPUs: 12 million parameters, 81.6% ImageNet top-1, 224×224 input resolution. EfficientNetV2 (Tan & Le, 2021) subsequently combined training-aware NAS with Fused-MBConv blocks for substantially faster training, though for the purposes of this study the original EfficientNet was sufficient.

Vision Transformers. The ViT (Dosovitskiy et al., 2021) reframed image classification as masked-sequence prediction over 16×16 image patches and demonstrated that, given sufficient pretraining data, plain transformers could match or exceed CNNs on standard benchmarks. The "given sufficient data" caveat is important: the original paper found that ViT only outperforms ResNet baselines when pretrained on either JFT-300M (Google's proprietary 300-million-image dataset) or ImageNet-21K (the public 14-million-image set); training from scratch on ImageNet-1K leaves ViT a step behind. This dependence on large-scale pretraining is the central reason ViT-based classifiers are typically used in fine-tuning rather than training-from-scratch regimes.

Several variants of ViT have proven important for downstream applications. Swin Transformer (Liu et al., 2021) introduced shifted-window self-attention, producing a hierarchical feature pyramid that integrates more naturally with dense prediction tasks and reaching 87.3% top-1 on ImageNet-1K. Hiera (Ryali et al., 2023) pushed in the opposite direction, removing the shifted windows and other architectural complexities and showing that a stripped-down hierarchical ViT, when paired with masked autoencoding pretraining, beats Swin on both accuracy and speed. This is the architecture inside SAM2's image encoder, which makes it indirectly relevant here as well.

Self-supervised ViT pretraining. The most consequential development for transfer learning has been the maturation of self-supervised pretraining for ViTs. DINO (Caron et al., 2021) showed that self-distillation without labels produces features that contain explicit semantic information — features that work, for instance, as strong k-NN classifiers with no further fine-tuning. DINOv2 (Oquab et al., 2024) scaled this idea to a 1-billion-parameter ViT distilled down to smaller, deployable models that outperform OpenCLIP on most image- and pixel-level benchmarks. The advertised use case for DINOv2 is precisely the setting one encounters in coral classification: limited labelled data, fine-grained classes, transfer from a large pretrained encoder. This is why DINOv2-S seemed, a priori, like a strong candidate for the classification backbone — a hypothesis

the experiments in Section 5 did not bear out.

Fine-grained classification. Coral species classification falls cleanly within the scope of *fine-grained image classification*: small inter-class visual variation (different species in the same genus look very similar), large intra-class variation (the same species varies in colour, growth form, and morphology across colonies and bleaching states). The comprehensive TPAMI survey by Wei et al. (Wei et al., 2022) taxonomises the field and notes that the challenges of fine-grained recognition — localising the discriminative parts of the object, learning subtle texture differences, dealing with viewpoint and illumination variation — recur across domains as varied as bird species recognition, vehicle classification, and species-level coral identification. The implication for this work is that coral classification is not a regular image classification problem and standard ImageNet-style training procedures may need adjustment.

2.4 Class imbalance and long-tailed learning

The species composition of any natural ecosystem is profoundly long-tailed: a small number of common species dominate while many rare species each contribute few observations. The CoralNet-derived dataset used in this study is no exception — *Madracis mirabilis* alone accounts for nearly a fifth of all annotations, while *Meandrina meandrites* contributes only 0.8%. This imbalance is a first-class concern for any deep classifier trained on this data, and there is a substantial methodological literature on how to handle it.

Loss reweighting. The most direct intervention is to modify the loss function so that rare classes receive proportionally more gradient signal. The two seminal references here are focal loss (Lin et al., 2017) and class-balanced loss based on effective number of samples (Cui et al., 2019). Focal loss adds a modulation term $(1 - p_t)^\gamma$ to standard cross-entropy, which downweights well-classified examples and concentrates learning on hard examples — this is most useful when the imbalance is between foreground and background but also helps with foreground-only imbalance. The Cui et al. formulation, on the other hand, observes that as one accumulates more samples from a class, the marginal information per sample decreases, and proposes weighting class c by $(1 - \beta)/(1 - \beta^{n_c})$ where n_c is the sample count and β is typically set close to one. The two losses can be combined, and on the long-tailed iNaturalist 2018 benchmark (the closest analog to a long-tailed coral species dataset I am aware of) this combination produces several percentage points of improvement over either alone.

Resampling. A second family of approaches modifies the data distribution rather than the loss. Weighted random sampling — the option I use in this report — draws training samples with probability inversely proportional to class frequency, ensuring that each minibatch contains a roughly uniform distribution of classes regardless of the underlying class frequencies in the dataset. This is the simplest possible intervention and frequently the most effective. The classical reference for synthetic minority oversampling, SMOTE (Chawla et al., 2002), predates deep learning by more than a decade but remains the canonical citation for any approach that generates synthetic minority examples. In the context of image classification, SMOTE is rarely used directly; standard augmentation (flips, crops, colour jitter) plays an analogous role.

Decoupled training and surveys. More recent work argues that the right solution depends

on the stage of training. Zhang et al. (Zhang et al., 2023b) provide a comprehensive TPAMI survey on deep long-tailed learning that distinguishes class re-balancing, information augmentation, and module improvement as three orthogonal axes; their meta-analysis suggests that decoupled training (where the feature extractor is trained on the natural distribution but the classifier is retrained on a balanced distribution) consistently outperforms single-stage methods. Fu et al. (Fu et al., 2022) provide a complementary survey organised by training stage and offers empirical benchmarks. Neither of these surveys focuses on biological imagery specifically, but the basic findings transfer well.

For the present work, the combination of weighted random sampling and focal loss with $\gamma = 2$ was sufficient to produce competitive results on the moderately-rare classes (*Pseudodiploria strigosa*, *Stephanocoenia intersepta*) but, as Section 5 shows, the very rare classes (*Orbicella franksi*, fewer than 500 training patches) remain stubbornly difficult. Future iterations of this work might profitably explore decoupled training in the style of Zhang et al.

2.5 Pre-foundation-model coral classification

For completeness, two earlier coral classification systems are worth mentioning. Gómez-Ríos et al. (Gómez-Ríos et al., 2019) benchmarked ResNet, Inception, and DenseNet variants with aggressive data augmentation on the EILAT and RSMAS coral texture datasets, showing that augmentation can substantially close the imbalance gap; their pipeline is essentially the pre-foundation-model template for what this report does. The CoralSeg system (Alonso et al., 2019), mentioned earlier, focused on converting sparse CoralNet point annotations into dense semantic segmentations through superpixel propagation, addressing a complementary problem.

More recent work has begun to bring SAM-style segmentation into the coral domain. Blondin et al. (Blondin et al., 2024) present a hierarchical classification scheme for Brazilian coral imagery, dealing explicitly with the long-tail problem (eleven out of fifty-four labels in their dataset account for 95% of annotations). Fawad et al. (Fawad et al., 2023) demonstrate that even hand-crafted feature pipelines can localise bleached corals competently, providing a useful contrast point for arguing that foundation-model approaches are not always strictly necessary.

3 Dataset and Annotation

The dataset assembled for this study consists of 19,152 underwater photoquadrat images sourced from eight publicly accessible CoralNet sources, yielding 66,219 species-level annotations across 17 Caribbean coral species (Table 1). All sources were downloaded through the CoralNet-Toolbox (Pierce, 2023), which automates the otherwise tedious process of fetching imagery and annotation CSVs from CoralNet sources one at a time.

Table 1: CoralNet sources contributing to the training and evaluation data.

Source	Location	Images	Annotations
Curaçao Coral Reef Assessment 2023	Curaçao	12,184	~43,000
Curaçao Reef Assessment 2025	Curaçao	1,494	~4,000
STINAPA GCRMN	Bonaire	1,713	~3,500
MarineGEO RLS	Multiple Caribbean	2,114	~8,000
LH CBC 2019–2022	Belize	443	~3,000
Little Cayman MarineGEO	Little Cayman	463	~1,500
FSU_STINAPA	Bonaire	665	~2,000
St. John 2016	St. John, USVI	78	~200
Total	Caribbean	19,152	66,219

3.1 Species mapping

Different CoralNet sources adopt different short-code conventions for the same species. The Curaçao 2023 source uses `OrbAnn` for *Orbicella annularis*, the LH CBC source uses `OANN`, and the FSU_STINAPA source uses `ORBANN`; each source has its own labelset stored in a per-source CSV. To consolidate these into a single label space, a mapping dictionary translates each source-specific code to one of 17 canonical species names. The dictionary also folds bleached variants (e.g. `BL_OrbAnn`) into the same species as healthy colonies on the rationale that they share morphology, even if colour is altered. This is a defensible choice given the goal of *species* identification, but it does mean the model is implicitly being asked to be invariant to bleaching state — an interesting topic that I do not explore further here.

After mapping, 66,219 annotations remained, distributed extremely unevenly across the 17 species (Table 2). The most common species (*Madracis mirabilis*) has nearly twenty times the annotation count of the rarest (*Meandrina meandrites*), and the test set, after a 70/15/15 image-level split, reflects this imbalance with 1,479 test samples for *M. mirabilis* versus 73 for *M. meandrites*. This ratio is broadly consistent with what one observes in real reef communities — common species are common — but it does create persistent measurement difficulties for the rare classes, as the test-set statistics for those species are computed from small samples and are correspondingly noisy.

Table 2: Species annotation counts and test set sizes after image-level 70/15/15 splitting.

Species	Annotations	Test samples	Frequency
<i>Madracis mirabilis</i>	10,263	1,479	15.5%
<i>Agaricia agaricites</i>	9,946	1,371	15.0%
<i>Orbicella annularis</i>	8,098	1,385	12.2%
<i>Orbicella faveolata</i>	7,348	1,209	11.1%
<i>Lobophyllia</i> spp	6,985	1,072	10.5%
<i>Millepora</i> spp	3,712	542	5.6%
<i>Porites astreoides</i>	3,695	593	5.6%
<i>Montastraea cavernosa</i>	3,468	479	5.2%
<i>Siderastrea siderea</i>	2,670	408	4.0%
<i>Colpophyllia natans</i>	2,359	322	3.6%
<i>Madracis auretenra</i>	1,924	264	2.9%
<i>Pseudodiploria strigosa</i>	1,385	208	2.1%
<i>Acropora tenuifolia</i>	1,329	130	2.0%
<i>Stephanocoenia intersepta</i>	1,044	143	1.6%
<i>Porites porites</i>	826	111	1.2%
<i>Orbicella franksi</i>	622	90	0.9%
<i>Meandrina meandrites</i>	545	73	0.8%
Total	66,219	9,879	100.0%

3.2 Train, validation, and test splits

Splits were assigned at the *image* level rather than at the *annotation* level: every annotation from the same image goes into the same split. This matters more than it might initially appear. Annotations within a single underwater photograph share lighting conditions, depth, water clarity, viewing angle, and often colony identity (annotations frequently point to nearby parts of the same colony). A randomly chosen train/test split at the annotation level would let the model memorise image-specific cues and report inflated test-time performance — a form of spatial leakage that has been documented repeatedly in the ecological remote-sensing literature (Wang et al., 2025). The 70/15/15 image-level split used here ensures that the test set contains only images the model has never seen, preserving the validity of the test-set metrics as an estimate of generalisation performance.

4 Methodology

The pipeline consists of two sequential stages: a segmentation stage that proposes coral colony masks, and a classification stage that assigns a species label to each mask. This section describes both, along with the patch-generation procedure used to assemble training data for the classifier.

4.1 Pipeline overview

At inference time, the pipeline operates as follows. Given an input image, SAM2’s automatic mask generator proposes a set of candidate object masks. Each mask is processed into a *masked crop* — a square bounding-box crop with the pixels outside the mask set to zero — and passed to EfficientNet-B3, which produces a 17-class probability distribution. Masks where the top-class confidence falls below a threshold (default 0.45) are rejected as likely non-coral. The remaining masks are rendered as a coloured species overlay along with a per-colony JSON output containing

species predictions, bounding boxes, and confidences.

At training time, the same SAM2 model is used to generate patches *from CoralNet annotation points* rather than from automatic mask proposals. This subtle change — which I will return to in Section 4.3 — is what allows the patches used to train the classifier to inherit the species labels that CoralNet annotators have already provided, with no additional manual effort.

4.2 Segmentation: SAM2

The SAM2 image encoder used in this work is `sam2.1_hiera_base_plus`, a 308 MB checkpoint with the Hiera-B+ backbone (Ryali et al., 2023). This is the second-smallest of the SAM2 model sizes; the choice is dictated primarily by GPU memory limits on the 8 GB consumer GPU used for development, but the larger `hiera_large` checkpoint would in principle be a drop-in upgrade if more memory were available.

For *inference*, the model is used in automatic-mask-generator mode with parameters tuned for coral imagery: `points_per_side = 32`, `pred_iou_thresh = 0.72`, `stability_score_thresh = 0.86`, `crop_n_layers = 1`, `crop_overlap_ratio = 0.45`, and `min_mask_region_area = 300`. These settings were arrived at iteratively, starting from the defaults and observing the behaviour on a small set of representative Curaçao photoquadrats. The defaults produce very many small masks (typically 200–400 per image), of which most are either substrate fragments or oversegmented portions of a single coral colony. The tuned settings produce roughly 50–100 masks per image, of which 5–15 typically pass the not-coral rejection threshold.

Several SAM2 design choices are worth flagging because they affect the quality of the segmentations in ways that are not always obvious. First, SAM2’s automatic mode samples a regular grid of foreground point prompts and runs them all through the model; the per-image time scales approximately linearly with the number of grid points, and the default 32-per-side gives 1,024 candidate prompts per image (further multiplied by cropping). Second, the `stability_score` measures how much the mask changes when the underlying threshold is perturbed — low stability scores indicate ambiguous boundaries, which on coral imagery typically correspond either to the edges of large colonies or to fragments of texture inside a larger colony. Third, the default mask post-processing includes a non-maximum suppression step that retains only one mask per area; on encrusting colonies with internal texture, this sometimes removes useful sub-colony masks but more often correctly collapses oversegmentations into a single colony.

4.3 Patch generation with SAM2 point prompts

This is the central methodological contribution of the report. The standard approach to generating training data for a coral species classifier would be to use SAM2’s automatic mask generator on each training image, then either match the resulting masks to CoralNet annotations spatially or, more commonly, hand-label each mask with its species. Both options have problems: automatic mask generation produces many false positives (substrate, algae, sand), and hand-labelling tens of thousands of masks is prohibitively expensive.

The approach I take instead exploits the fact that CoralNet annotations are already given as (row, column, species) triples (Beijbom et al., 2015). For each annotation, I feed the (column,

row) coordinate to SAM2 as a foreground point prompt and request three candidate masks (the standard `multimask_output = True` setting). I select the mask that (a) contains the annotation point and (b) has the highest predicted IoU score. This mask is then converted to a masked crop and saved with the species label inherited from the annotation. Because the same image is reused for many annotation points, the SAM2 image encoder is run only once per image and its embedding cached for all subsequent prompts on that image; this drops the amortised cost from roughly four seconds per annotation to about 0.3 seconds.

The result is a training set in which every patch is guaranteed to contain a confirmed coral colony with a known species. There are no false positives. The mask quality is also somewhat better than would be obtained from automatic mode, because SAM2 with a single foreground point tends to produce the most coherent single-object mask (in contrast to automatic mode, which produces all plausible object decompositions). Some patches do fail — if the mask area falls below 100 pixels, the patch is rejected as too small — but the failure rate is low (0.04% in this study).

In total, 43,672 patches were extracted across the 17 species and three splits, with the resulting per-species distribution given in Table 6. The total drops from the 66,219 input annotations primarily because of the 100-pixel minimum area threshold; some morphologies (very small foliate colonies, thin branching *Acropora*) yield masks below this size more frequently than others.

4.4 Classification: EfficientNet-B3

The classification backbone is EfficientNet-B3 (Tan & Le, 2019), pretrained on ImageNet-1K and loaded through the `timm` library. The 1000-class ImageNet output head is replaced with a 17-class linear layer, and the model is fine-tuned in two phases:

Phase 1 (3 epochs). The backbone is frozen and only the new classification head is trained, at learning rate 10^{-3} with AdamW. The purpose is to stabilise the randomly initialised head before any gradient flows back through the pretrained backbone — without this warm-up, the large gradients produced by a randomly initialised head can destabilise the carefully tuned backbone weights and cause a measurable loss of accuracy.

Phase 2 (up to 57 epochs). All layers are unfrozen. The backbone learns at 10^{-4} while the classification head continues at 10^{-3} ; this disparity reflects the standard intuition that pretrained backbones should be perturbed gently while task-specific heads should be allowed to adapt freely. Learning rates are decayed by cosine annealing to 1% of their initial values over the full schedule. Early stopping with patience 12 terminates training if the validation F1 score does not improve.

Loss. The composite loss combines focal loss (Lin et al., 2017) with $\gamma = 2$ and label smoothing of 0.05. Focal loss is preferred over plain cross-entropy because it downweights well-classified examples and concentrates gradient signal on the hard ones — a useful property given the underlying class imbalance and the within-class variability of coral appearance. Label smoothing prevents the model from becoming overconfident, which is helpful for downstream confidence-thresholded rejection.

Class balance. A weighted random sampler ensures that each minibatch contains a roughly uniform distribution of species, regardless of the underlying class frequencies. This is the simplest

form of class rebalancing and was sufficient for the moderate imbalance in this dataset; more sophisticated approaches such as effective-number-of-samples reweighting (Cui et al., 2019) or decoupled training (Zhang et al., 2023b) could be explored in future work.

Augmentation. The augmentation pipeline includes `RandomResizedCrop` to 224 pixels, horizontal and vertical flips, 30° rotation, colour jitter (brightness, contrast, saturation), Gaussian blur, and random erasing. For the four lowest-support classes (*Meandrina meandrites*, *Orbicella franksi*, *Porites porites*, *Stephanocoenia intersepta*), offline augmentation generates five copies of each training patch by flips and rotations, raising the effective training set size for these classes.

Optimisation. AdamW with weight decay 3×10^{-4} , batch size 32, mixed-precision training, with gradient clipping at norm 1.0. Weight decay was tuned by sweeping between 10^{-4} and 5×10^{-4} ; the higher value gave marginally better held-out performance, presumably because the larger batch size and the diversity of the augmentations reduces the risk of overfitting.

5 Experiments and Results

This section reports the main experimental results. Three ablations are presented: the effect of switching the segmentation model from MobileSAM to SAM2, the effect of switching the classification backbone from EfficientNet-B3 to DINOv2-S, and the per-species breakdown of the final model. All experiments use the same image-level train/validation/test splits described in Section 3.

5.1 Segmentation model: MobileSAM versus SAM2

The first ablation compares MobileSAM and SAM2 as the segmentation model used for patch generation, holding the classifier architecture (EfficientNet-B3) and all training hyperparameters constant.

Table 3: Effect of segmentation model on classification performance. All rows use EfficientNet-B3 fine-tuned on masked crops.

Segmenter	Weight decay	Patches	Accuracy	Macro-F1	Val F1
MobileSAM	10^{-4}	49,850	69.5%	0.603	0.635
MobileSAM	3×10^{-4}	49,850	70.1%	0.606	0.635
SAM2	3×10^{-4}	43,672	72.6%	0.624	0.626

The SAM2 model trained on 6,178 *fewer* patches outperforms the best MobileSAM-trained model by 0.018 macro-F1 and 2.5 percentage points of accuracy. The improvement is statistically meaningful given the test set size (9,879 samples); a one-sided binomial test on accuracy gives $p < 10^{-4}$. The finding is consistent with the central insight of Wang et al. (Wang et al., 2025) that data *quality* dominates data *quantity*: when SAM2 rejects a marginal mask that MobileSAM would have kept, the classifier is better off training on the remaining cleaner examples. The mechanism is straightforward: cleaner colony masks mean less background contamination in the masked crops, and less background means the classifier can focus its capacity on actual coral texture rather than the random rubble visible at the mask boundary.

5.2 Classification backbone: EfficientNet-B3 versus DINOv2-S

The second ablation holds the patch generation method fixed (SAM2 point prompts, masked crops) and compares two classification backbones. EfficientNet-B3 is the chosen baseline; DINOv2-S (Oquab et al., 2024) is a self-supervised Vision Transformer that, on paper, should be well-suited to fine-grained classification on limited labelled data.

Table 4: Effect of classification backbone on SAM2 masked crops.

Backbone	Parameters	Accuracy	Macro-F1	Train/Val Acc
DINOv2-S	22 M	46.8%	0.376	80% / 44%
EfficientNet-B3	12 M	72.6%	0.624	97% / 72%

The DINOv2-S result is striking and, frankly, not what I had expected when I set up the experiment. The 25 percentage point gap in test accuracy is large, and the train/validation accuracy split (80%/44%) indicates catastrophic overfitting — the model memorises training patches but fails to generalise. EfficientNet-B3 on the same data shows much more modest overfitting (97%/72%) and the test set tracks the validation set closely.

The most likely explanation, which I develop in more detail in Section 6, is a distribution-shift problem. DINOv2’s self-supervised pretraining (Oquab et al., 2024) uses a contrastive objective that asks the model to produce consistent features for different crops of the same image, drawn from a corpus of 142 million natural images. The masked crops used here contain 25–40% zeroed-out background pixels, which are not part of the pretraining distribution; the model’s patch-level attention mechanism is forced to attend to these black regions and the resulting features are degraded. This is consistent with the broader observation in the DINOv2 paper that the model’s features depend heavily on the patches being natural-image-like; it is also consistent with reports from the SAM-medical literature that ViT models adapt poorly when input statistics differ substantially from natural images (Ma et al., 2024). EfficientNet-B3, with its convolutional inductive bias, is more tolerant of the abrupt boundaries introduced by masking; the convolutions can effectively "see through" the zeroed regions because they aggregate over local neighbourhoods rather than attending globally.

This result is not a general claim about ViT backbones being inferior to CNN backbones for coral classification — it is specifically a result about ViT backbones on masked crops with zeroed backgrounds. The natural next experiment would be to retrain DINOv2-S on bounding-box crops that preserve the natural background and see whether the gap closes. That experiment is left for future work.

5.3 Final model: per-species results

The complete per-species results for the final model are given in Table 5.

Table 5: Per-species classification results on the held-out test set ($n = 9,879$).

Species	Precision	Recall	F1	Support
<i>Madracis mirabilis</i>	0.911	0.855	0.882	1,479
<i>Orbicella annularis</i>	0.834	0.757	0.794	1,385
<i>Siderastrea siderea</i>	0.727	0.828	0.774	408
<i>Agaricia agaricites</i>	0.826	0.718	0.768	1,371
<i>Lobophyllia</i> spp	0.737	0.730	0.734	1,072
<i>Porites astreoides</i>	0.697	0.734	0.715	593
<i>Colpophyllia natans</i>	0.736	0.683	0.709	322
<i>Orbicella faveolata</i>	0.739	0.669	0.703	1,209
<i>Montastraea cavernosa</i>	0.668	0.722	0.694	479
<i>Millepora</i> spp	0.676	0.690	0.683	542
<i>Pseudodiploria strigosa</i>	0.542	0.625	0.580	208
<i>Acropora tenuifolia</i>	0.441	0.638	0.522	130
<i>Madracis auretenra</i>	0.527	0.481	0.503	264
<i>Stephanocoenia intersepta</i>	0.396	0.559	0.464	143
<i>Meandrina meandrites</i>	0.336	0.671	0.447	73
<i>Porites porites</i>	0.330	0.622	0.431	111
<i>Orbicella franksi</i>	0.145	0.322	<i>0.200</i>	90
Macro average	0.604	0.665	0.624	—
Weighted average	0.748	0.726	0.734	—

The performance distribution is bimodal. Eleven species achieve F1 scores above 0.60, with the top four exceeding 0.75. The remaining six species cluster between 0.20 and 0.55, and the worst performer (*Orbicella franksi*, $F1 = 0.200$) is essentially unusable for downstream ecological analysis at this confidence level. There is a clear correlation between training-set size and F1: the linear regression of F1 on $\log(\text{support})$ yields $R^2 = 0.78$, with the rare species systematically falling below the regression line. There are also visible morphology effects independent of sample size — *Meandrina meandrites* (foliate, distinctive ridge pattern) outperforms its training-count prediction, while *Madracis auretenra* (very fine branching, easily confused with congeners) underperforms.

5.4 Patch-count comparison

For completeness, Table 6 compares the number of training patches generated per species by MobileSAM and SAM2. The aggregate numbers are similar (49,850 vs. 43,672), but there are large per-species differences worth noting.

Table 6: Training patch counts per species: MobileSAM versus SAM2.

Species	MobileSAM	SAM2	Change
<i>Madracis mirabilis</i>	7,017	7,226	+209
<i>Agaricia agaricites</i>	6,283	7,046	+763
<i>Orbicella annularis</i>	5,204	5,424	+220
<i>Orbicella faveolata</i>	4,659	5,112	+453
<i>Lobophyllia</i> spp	4,490	4,883	+393
<i>Millepora</i> spp	2,390	2,633	+243
<i>Porites astreoides</i>	2,369	2,603	+234
<i>Montastraea cavernosa</i>	2,227	2,430	+203
<i>Siderastrea siderea</i>	1,714	1,870	+156
<i>Colpophyllia natans</i>	1,514	1,669	+155
<i>Madracis auretenra</i>	1,240	1,372	+132
<i>Pseudodiploria strigosa</i>	882	948	+66
<i>Acropora tenuifolia</i>	1,364	976	-388
<i>Stephanocoenia intersepta</i>	622	767	+145
<i>Porites porites</i>	529	633	+104
<i>Orbicella franksi</i>	391	441	+50
<i>Meandrina meandrites</i>	865	378	-487
Total	43,760	44,405	+645

Two species see substantial losses with SAM2: *Meandrina meandrites* drops by 487 patches (56%), and *Acropora tenuifolia* drops by 388 patches (28%). Both species share a morphological pattern — small, thin, or finely branched colonies — that produces mask areas below the 100-pixel rejection threshold more frequently than other species. This is the most likely explanation for these species’ relatively poor F1 scores: not just that they have few annotations, but that a disproportionate fraction of those annotations produce no usable patch at all. Future work might experiment with a lower area threshold or species-specific thresholds for known small-colony species.

6 Discussion

This section interprets the experimental results, places them in the context of the broader literature, and identifies the limitations that should constrain how the pipeline is used and what should be done next.

6.1 Why SAM2 helps

The 0.018 macro-F1 improvement from switching to SAM2 is consistent with what one would predict from the architectural differences between MobileSAM and SAM2. MobileSAM (Zhang et al., 2023a) was designed to be small and fast, not accurate; the 5-million-parameter Tiny-ViT image encoder is approximately one-hundredth the size of the original SAM’s ViT-H, and the distillation procedure inevitably loses some boundary information. SAM2’s Hiera backbone (Ryali et al., 2023) is much heavier and was trained with a richer masked-autoencoder pretraining objective, producing features that retain finer-grained spatial information. The downstream consequence is sharper mask boundaries on coral colonies, particularly on the textured surfaces of encrusting species and the branched structure of branching species.

The improvement is most dramatic for *Millepora* (fire coral), which gained roughly 0.19 F1 with

the SAM2 patches. *Millepora* has a complex branching morphology that MobileSAM’s smoother boundaries failed to capture — the resulting patches often had ragged edges that included background coral or substrate. SAM2 produces masks that are tightly fitted to the branching structure, which means the masked crops are nearly pure *Millepora*. The classifier benefits accordingly. The smallest improvements are on species where boundary precision matters least — the large, smooth *Madracis mirabilis* colonies that are already easy targets for any segmentation model. This pattern aligns with the general finding from HQ-SAM (Ke et al., 2023) that the marginal value of improved boundary precision depends heavily on the structural complexity of the target.

6.2 Why DINOv2 fails (and what to do about it)

The DINOv2-S result deserves more discussion because it is, on its face, surprising. DINOv2 (Oquab et al., 2024) reports state-of-the-art transfer learning performance across more than twenty image-level and pixel-level benchmarks, and the self-supervised objective is precisely the kind that should produce features useful for fine-grained classification. Why does it fail here?

The most defensible answer is that the masked crops sit outside the input distribution under which DINOv2 was pretrained. DINOv2’s training procedure operates on natural images, with crops that share the local statistics of those images. A typical masked crop, in contrast, has 25–40% of its pixels set to exactly zero — a discontinuity that does not appear in any natural image. The transformer’s self-attention mechanism processes the image as a sequence of 14×14 patches; many of these patches are entirely zero, and the model has no good prior for what to do with them. The output features for these zeroed patches degrade the global representation, particularly when the attention layers aggregate features across the full image.

EfficientNet-B3 is more tolerant of this kind of perturbation because of the locality of convolution. A convolutional kernel that straddles the mask boundary aggregates a few "real" pixels with a few zeros; the resulting feature is dampened but not catastrophically wrong, and the next layer of convolutions can recover. The cumulative effect over the network is something close to a brightness shift on the masked regions, which is a perturbation EfficientNet has presumably seen many times during its ImageNet pretraining. The transformer’s global attention, by contrast, can be derailed by anomalous patches in a way that local convolution cannot.

If this analysis is correct, two interventions should help. First, training DINOv2 on bounding-box crops that preserve natural background should eliminate the distribution-shift problem entirely; the model would then see only natural underwater patches. Second, applying an attention mask that prevents the transformer from attending to zeroed patches would solve the problem on masked crops directly. The second intervention requires modifying the model’s forward pass and is more invasive; the first is a drop-in change and would be the natural first experiment.

This finding is also consistent with results in the medical imaging literature on SAM adaptation. Ma et al. (Ma et al., 2024), in their large-scale MedSAM study, observed that SAM-style ViT encoders adapt poorly to medical imagery without explicit domain adaptation; the lesson there is that ViT pretraining is brittle to input distribution shifts in a way that is not always apparent until one tries to apply the model. The DINOv2 result reported here is, in some sense, a coral-domain analogue of that finding.

6.3 Persistent limitations

Class imbalance. The rare species (*O. franksi*, *M. meandrites*, *P. porites*) remain weakly classified despite the use of weighted random sampling and focal loss. This is not a surprise — the long-tailed-learning literature (Zhang et al., 2023b; Fu et al., 2022) suggests that these single-stage rebalancing techniques close most but not all of the imbalance gap, and the residual gap grows with the imbalance ratio. The imbalance ratio in this dataset is roughly 20:1 between the most and least common species, which puts it in the "moderately imbalanced" regime where decoupled training (Zhang et al., 2023b) typically helps. The natural next step would be a two-stage training procedure in which the feature extractor is trained on the natural distribution but the classification head is retrained on a balanced distribution. This was not implemented in the current pipeline but is a clear candidate for future work.

Morphology-specific under-segmentation. The 100-pixel minimum mask area filters out small-colony annotations for *Meandrina meandrites* and *Acropora tenuifolia* at a much higher rate than for other species. There are at least three ways to address this. The simplest is to lower the area threshold globally, accepting more noise in exchange for higher coverage of small species. A more principled approach is species-specific thresholds, set based on observed minimum mask sizes per species. The cleanest fix, and the one I would recommend if compute budget allows, is to retrain SAM2 (or use a fine-tuned variant such as CoralSCOP (Zheng et al., 2024a)) on coral-specific data, which should improve the small-mask quality enough that the threshold can be lowered without introducing too many spurious matches.

Colony fragmentation. SAM2’s automatic mode occasionally segments large encrusting colonies (*O. annularis*, *Colpophyllia natans*) into multiple sub-masks corresponding to individual corallite bumps or growth ridges. This was less of a problem for the patch-extraction step (where each mask is just one training example among many) than for inference time, where the user-visible output shows a single colony split into several differently coloured pieces. A post-processing pass that merges adjacent same-species masks within a distance threshold would resolve this; preliminary experiments with this approach showed it works but introduces its own failure mode (it occasionally merges genuinely separate colonies of the same species growing close together). I leave this as future work.

False positives. Non-coral substrate — sand, dead coral, sponges, encrusting algae — is occasionally segmented and misclassified as coral. The not-coral rejection threshold (default 0.45 on the top-class confidence) catches most of these, but the threshold is necessarily a global value chosen by trial and error, and it produces both false positives (substrate confidently misclassified) and false negatives (genuine but uncertain coral rejected). A more principled approach would be to add an explicit *not-coral* class to the classifier and train it with negative examples drawn from CoralNet’s substrate annotations. This is well within the standard fine-grained classification literature (Wei et al., 2022) and is another clear candidate for future work.

6.4 Comparison to published baselines

Direct comparison with the Coral-Lab benchmark (Wang et al., 2025) requires some care because the tasks are not identical. Coral-Lab achieves F-score 0.802 and mIoU 0.665 on *binary* coral-

versus-non-coral segmentation; the present work achieves macro-F1 0.624 across *seventeen* coral species. The 17-class problem is fundamentally harder because it requires the model to distinguish congeneric species that share most of their visual features. A more apples-to-apples comparison would aggregate the 17 species predictions in this study into a binary coral-versus-background mask, in which case the implied accuracy would be substantially higher (since most species-level errors are confusions between similar species, not confusions between coral and non-coral). I have not run this analysis but it would be a useful sanity check.

The CoralSCOP benchmark (Zheng et al., 2024a) reports pixel accuracy around 86% on unseen reef images for the binary coral-segmentation task. Again, the species-level numbers in the present work are not directly comparable, but the implicit binary accuracy is in the same ballpark.

Within the species-level coral classification literature, the most directly comparable result I am aware of is the Brazilian hierarchical classification work of Blondin et al. (Blondin et al., 2024), which reports a similar macro-F1 in the 0.5–0.6 range on a 54-label dataset where 11 labels account for 95% of annotations. Their dataset is substantially smaller (1,549 images), which makes their numbers if anything more impressive than mine for a fair direct comparison.

7 Conclusions and Future Work

This report has presented an end-to-end pipeline for species-level classification of Caribbean coral reef imagery, combining SAM2 for instance segmentation with a fine-tuned EfficientNet-B3 for species identification across 17 species. The final model achieves macro-F1 = 0.624 and accuracy = 72.6% on a held-out test set of 9,879 patches drawn from images the model has never seen. The pipeline is fully implemented and deployable on consumer hardware (an 8 GB GPU). All code is publicly released at <https://github.com/ARLab2018/Caribbean-Coral-Classification> under the MIT licence, with the trained model checkpoints, source images, and generated patches hosted on Google Drive and linked from the repository README. The repository contains the three patch extractors used in this study (MobileSAM, CoralSCOP, SAM2), the unified training script (which supports both EfficientNet-B3 and DINOv2-S backbones), the four inference scripts corresponding to the four end-to-end configurations evaluated, and the manual labelling tool described in Section 4.

The main empirical findings are three. First, SAM2 with point-prompt patch generation outperforms a comparably trained MobileSAM baseline by 0.018 macro-F1 despite training on roughly 6,000 fewer patches; the improvement is most pronounced on species with complex branching or textured morphology. Second, a DINOv2-S Vision Transformer fails badly on the same masked crops — macro-F1 drops to 0.376 with severe overfitting — which I attribute to a distribution mismatch between the zeroed background of masked crops and the natural-image patches DINOv2 was pretrained on. Third, the persistent classification errors concentrate on rare species (*O. franksi*, *M. meandrites*, *P. porites*) that combine low annotation counts with visual similarity to more common congeners.

7.1 Recommendations for follow-on work

If I were continuing this work, I would prioritise the following experiments in roughly the listed order:

- 1. Retrain DINOv2 on natural-background crops.** This is the most direct way to test the distribution-shift hypothesis for the ViT failure. If DINOv2 on bounding-box crops with natural background matches or exceeds EfficientNet-B3 on masked crops, that would be a useful positive result and would suggest that ViTs are viable for this task with the right input. If DINOv2 still underperforms, the explanation must lie elsewhere (perhaps in the fine-grained discrimination capacity of self-supervised features for closely related coral species), which would itself be informative.
- 2. Drop in CoralSCOP for segmentation.** Replacing SAM2 with CoralSCOP (Zheng et al., 2024a) would test whether coral-domain pretraining improves segmentation enough to translate into downstream classification gains. CoralSCOP’s authors report substantially fewer over-segmentation artefacts than vanilla SAM, which suggests it would help with the colony-fragmentation issue I noted in Section 6. The patch-extraction pipeline would need to be adapted to use CoralSCOP’s prompt API rather than SAM2’s, but the core logic transfers cleanly.
- 3. Add a not-coral class with hard negative mining.** The current confidence-threshold rejection is brittle and global. An explicit not-coral class trained with substrate, algae, sand, and sponge patches sampled from the same CoralNet sources would give a more principled false-positive rejection. The not-coral patches could be generated by sampling random points outside the coral annotations and running them through the same SAM2 point-prompt patch extractor.
- 4. Decoupled training for the rare classes.** Train the feature extractor on the natural class distribution and retrain only the classifier head on a class-balanced distribution (Zhang et al., 2023b). This is a small modification to the existing training pipeline and the long-tailed-learning survey literature suggests it consistently improves performance on the tail of the class distribution.
- 5. Annotate 50–100 additional images per rare species.** The hardest cases are bottlenecked by data, not by architecture. The label tool included in the code repository allows new images to be annotated manually; for the three weakest species, a focused annotation effort would likely improve their F1 scores more than any architectural change.
- 6. Expand the label set to non-coral benthic categories.** For full benthic-cover mapping (the end-use case for many ecological monitoring programs), the pipeline needs to also classify crustose coralline algae (CCA), macroalgae, sponges, rubble, and sand. Most of these categories are already labelled in CoralNet sources, so the labelset can be extended without new fieldwork.

7.2 Closing remarks

Foundation models have changed what is achievable in segmentation, and the lift here — moving from a hand-built U-Net to a SAM-based pipeline — is real but not magical. The model is competent on common species, struggles on rare ones, and has obvious failure modes that the next iteration of the work should address. What I think is encouraging is that the bottleneck has shifted from segmentation quality (where SAM2 and CoralSCOP are now strong enough not to be

the limiting factor) to data quantity for rare species and to the choice of classification backbone. Both of those bottlenecks are more tractable than the segmentation bottleneck used to be, and I expect a future version of this work, with the modifications above, to reach macro-F1 in the 0.70–0.75 range without any further foundation-model improvements.

Code and Data Availability

The full source code for this project is released under the MIT licence at:

<https://github.com/ARLab2018/Caribbean-Coral-Classification>

The repository contains the three patch-extraction scripts (one per segmentation model evaluated — MobileSAM, CoralSCOP, SAM2), the unified training script that produces both the EfficientNet-B3 and DINOv2-S models reported here, four inference scripts corresponding to the four end-to-end configurations, the manual labelling tool, and a copy of this report along with its LaTeX source.

Large artefacts that cannot be hosted on GitHub — the four trained model checkpoints (approximately 600 MB combined), the 19,152 source images downloaded from the CoralNet sources (approximately 12 GB), and the 43,672 generated SAM2 training patches (approximately 5 GB) — are hosted on Google Drive with links provided in the repository’s README. The SAM2 base weights are obtained directly from Meta AI Research; MobileSAM weights from the MobileSAM repository; CoralSCOP weights from the CoralSCOP project page.

Licensing note. The code is MIT-licensed and may be used freely, including commercially. The trained model weights, however, were produced from CoralNet annotations distributed under CC BY-NC 4.0 and therefore inherit a non-commercial restriction. Commercial deployment of the trained models would require either retraining on data with a permissive licence or obtaining explicit permission from the original CoralNet contributors.

Acknowledgements

This work used CoralNet data from eight publicly accessible sources and benefited from CoralNet-Toolbox by Jordan Pierce. The Coral-Lab paper by Wang et al. provided the primary benchmark reference. SAM2 weights and code were obtained from Meta AI Research’s open-source releases. I thank my supervisor for feedback throughout the project and for suggesting the comparison with foundation models that ultimately motivated the SAM2 line of work.

References

- Fawad, Ahmad, I., Ullah, A., & Choi, W. (2023). Machine learning framework for precise localization of bleached corals using bag-of-hybrid visual feature classification. *Scientific Reports*, 13:19461. DOI: 10.1038/s41598-023-46971-7.
- Alonso, I., Yuval, M., Eyal, G., Treibitz, T., & Murillo, A. C. (2019). CoralSeg: Learning coral segmentation from sparse annotations. *Journal of Field Robotics*, 36(8):1456–1477. DOI: 10.1002/rob.21915.

- Beijbom, O., Edmunds, P. J., Roelfsema, C., et al. (2015). Towards automated annotation of benthic survey images: Variability of human experts and operational modes of automation. *PLOS ONE*, 10(7):e0130312. DOI: 10.1371/journal.pone.0130312.
- Caron, M., Touvron, H., Misra, I., Jégou, H., Mairal, J., Bojanowski, P., & Joulin, A. (2021). Emerging properties in self-supervised vision transformers. *Proceedings of ICCV 2021*. arXiv:2104.14294.
- Chawla, N. V., Bowyer, K. W., Hall, L. O., & Kegelmeyer, W. P. (2002). SMOTE: Synthetic minority over-sampling technique. *Journal of Artificial Intelligence Research*, 16:321–357. DOI: 10.1613/jair.953.
- Cui, Y., Jia, M., Lin, T.-Y., Song, Y., & Belongie, S. (2019). Class-balanced loss based on effective number of samples. *Proceedings of CVPR 2019*, pp. 9268–9277. arXiv:1901.05555.
- Dosovitskiy, A., Beyer, L., Kolesnikov, A., et al. (2021). An image is worth 16×16 words: Transformers for image recognition at scale. *Proceedings of ICLR 2021*. arXiv:2010.11929.
- Eddy, T. D., Lam, V. W. Y., Reygondeau, G., et al. (2021). Global decline in capacity of coral reefs to provide ecosystem services. *One Earth*, 4(9):1278–1285. DOI: 10.1016/j.oneear.2021.08.016.
- Fu, Y., Xiang, L., Zahid, Y., et al. (2022). Long-tailed visual recognition with deep models: A methodological survey and evaluation. *Neurocomputing*, 509:290–309. DOI: 10.1016/j.neucom.2022.08.031.
- Gómez-Ríos, A., Tabik, S., Luengo, J., et al. (2019). Towards highly accurate coral texture images classification using deep convolutional neural networks and data augmentation. *Expert Systems with Applications*, 118:315–328. arXiv:1804.00516.
- He, K., Zhang, X., Ren, S., & Sun, J. (2016). Deep residual learning for image recognition. *Proceedings of CVPR 2016*, pp. 770–778. DOI: 10.1109/CVPR.2016.90.
- Jackson, J. B. C., Donovan, M. K., Cramer, K. L., & Lam, V. V. (eds.) (2014). *Status and Trends of Caribbean Coral Reefs: 1970–2012*. Global Coral Reef Monitoring Network, IUCN, Gland, Switzerland.
- Ke, L., Ye, M., Danelljan, M., Liu, Y., Tai, Y.-W., Tang, C.-K., & Yu, F. (2023). Segment anything in high quality. *NeurIPS 2023*. arXiv:2306.01567.
- Kirillov, A., Mintun, E., Ravi, N., et al. (2023). Segment anything. *Proceedings of ICCV 2023*, pp. 4015–4026. DOI: 10.1109/ICCV51070.2023.00371. arXiv:2304.02643.
- Lian, S., & Li, H. (2024). Evaluation of Segment Anything Model 2: The role of SAM2 in the underwater environment. arXiv:2408.02924.
- Lian, S., Zhang, Z., Li, H., et al. (2024). Diving into underwater: Segment Anything Model guided underwater salient instance segmentation and a large-scale dataset. *Proceedings of ICML 2024*, PMLR 235:29545–29559. arXiv:2406.06039.
- Lin, T.-Y., Goyal, P., Girshick, R., He, K., & Dollár, P. (2017). Focal loss for dense object detection. *Proceedings of ICCV 2017*, pp. 2980–2988. arXiv:1708.02002.

- Liu, Z., Lin, Y., Cao, Y., et al. (2021). Swin Transformer: Hierarchical Vision Transformer using shifted windows. *Proceedings of ICCV 2021*, pp. 10012–10022. arXiv:2103.14030.
- Ma, J., He, Y., Li, F., Han, L., You, C., & Wang, B. (2024). Segment anything in medical images. *Nature Communications*, 15:654. DOI: 10.1038/s41467-024-44824-z.
- Oquab, M., Darcet, T., Moutakanni, T., et al. (2024). DINOv2: Learning robust visual features without supervision. *Transactions on Machine Learning Research*. arXiv:2304.07193.
- Pierce, J. (2023). *CoralNet-Toolbox*. NOAA. <https://github.com/Jordan-Pierce/CoralNet-Toolbox>
- Blondin, C., Guérin, J., Inagaki, K., Longo, G., & Berti-Équille, L. (2024). Hierarchical classification for automated image annotation of coral reef benthic structures. arXiv:2412.08228.
- Ravi, N., Gabeur, V., Hu, Y.-T., et al. (2024). SAM2: Segment anything in images and videos. arXiv:2408.00714.
- Ryali, C., Hu, Y.-T., Bolya, D., et al. (2023). Hiera: A hierarchical Vision Transformer without the bells-and-whistles. *Proceedings of ICML 2023* (Oral), pp. 29441–29454. arXiv:2306.00989.
- Tan, M., & Le, Q. V. (2019). EfficientNet: Rethinking model scaling for convolutional neural networks. *Proceedings of ICML 2019*, PMLR 97:6105–6114. arXiv:1905.11946.
- Tan, M., & Le, Q. V. (2021). EfficientNetV2: Smaller models and faster training. *Proceedings of ICML 2021*, PMLR 139:10096–10106. arXiv:2104.00298.
- Wang, J., Mizuno, K., Tabeta, S., et al. (2025). Multi-dataset-integrated Coral-Lab segmentation with enhanced towed camera array for rapid large-scale coral reef monitoring and mapping. *International Journal of Applied Earth Observation and Geoinformation*, 143:104819. DOI: 10.1016/j.jag.2025.104819.
- Wei, X.-S., Song, Y.-Z., Mac Aodha, O., Wu, J., Peng, Y., Tang, J., Yang, J., & Belongie, S. (2022). Fine-grained image analysis with deep learning: A survey. *IEEE TPAMI*, 44(12):8927–8948. arXiv:2111.06119.
- Xiong, Y., Varadarajan, B., Wu, L., et al. (2024). EfficientSAM: Leveraged masked image pretraining for efficient segment anything. *Proceedings of CVPR 2024*, pp. 16111–16121. arXiv:2312.00863.
- Zhang, C., Han, D., Qiao, Y., Kim, J. U., Bae, S. H., Lee, S., & Hong, C. S. (2023). Faster segment anything: Towards lightweight SAM for mobile applications. arXiv:2306.14289.
- Zhang, Y., Kang, B., Hooi, B., Yan, S., & Feng, J. (2023). Deep long-tailed learning: A survey. *IEEE TPAMI*, 45(9):10795–10816. arXiv:2110.04596.
- Zhao, X., Ding, W., An, Y., Du, Y., Yu, T., Li, M., Tang, M., & Wang, J. (2023). Fast segment anything. arXiv:2306.12156.
- Zheng, Z., Liang, H., Hua, B.-S., Wong, Y. H., Ang, P. Jr., Chui, A. P. Y., & Yeung, S.-K. (2024). CoralSCOP: Segment any COral image on this Planet. *Proceedings of CVPR 2024* (Highlight), pp. 28170–28180. DOI: 10.1109/CVPR52733.2024.02661.

- Wong, Y.-K., Zheng, Z., Zhang, M., Suggett, D., & Yeung, S.-K. (2024). CoralSCOP-LAT: Labeling and analyzing tool for coral reef images with dense mask. arXiv:2410.20436.
- Zheng, Z., Chen, Y., Zeng, H., Vu, T.-A., Hua, B.-S., & Yeung, S.-K. (2024). MarineInst: A foundation model for marine image analysis with instance visual description. *ECCV 2024* (Oral), LNCS 15060. DOI: 10.1007/978-3-031-72627-9_14.